Controlled Quantum-Weight Networks: Exceptional Pairing, Universal Proliferation and Collective Limits

29 October 2026

Abstract

A four-level quantum control cell can realize a compact symplectic algebra rather than the full special-unitary algebra. We study a positive-weight family whose two weighted logical controls and transverse update generate exactly $\mathfrak{spin}(5)$. The weight direction selects its six-plus-four symmetric decomposition and a polarized Clifford trace. A vector-polarization bond between two independently controlled cells generates $\mathfrak{spin}(10)$ on one chiral sixteen-dimensional module. This exceptional pairing fails to scale: every connected network of at least three independently controlled cells, with the specified vector or clock-bivector bonds independently available, generates the full $\mathfrak{su}(4^n)$. We prove the network extension by a finite Clifford-algebra induction, with exact integer Pauli calculations supplying the small-system bases. A separate iid collective limit gives three canonical pairs with an explicit polarization-noise bound. Its diagonal commutator error vanishes in state root mean square but has constant nonzero operator norm. The phase Hamiltonian is a transverse Curie–Weiss model with spectator degrees of freedom; it supplies compatible one-site polarizations but does not establish an interacting-Gibbs fluctuation theorem. The result is a finite control construction with a completed proliferation obstruction and precise collective-limit boundaries.

1 Questions and scope

Restricted orthogonal and symplectic dynamical algebras are established in quantum control [1, 2]. Clifford realizations make their representations explicit, while generic added interactions can restore full controllability. This makes an exceptional two-cell algebra scientifically useful only after its behavior under repetition has been determined. We ask three connected questions: which algebra follows from the positive-weight control family; whether its chiral pairing extends across a network; and which large-block variables survive if independent microscopic control is replaced by a specified product-state fluctuation limit.

The component group-theoretic identities are standard. The contributions are an explicit weight-parameterized control family and its update restriction, a complete connected-network classification for two bond types, and an exact distinction between state and operator-norm convergence in its collective construction. The local algebra is also a Euclidean Cartan carrier, and finite embedding algebras have a long history [3, 4, 5]. These motivations do not identify the present variables with a spacetime, a coframe, a matter field or gravitational dynamics. All controls in the network theorem are independently available real Hamiltonian directions; the theorem does not concern a single fixed homogeneous Hamiltonian.

2 A weighted four-level dynamical algebra

Let X, Y, Z denote the Pauli matrices and I the 2×2 identity. On $\mathcal{H} = \mathbb{C}^2_w \otimes \mathbb{C}^2_\ell$ define

$$G = \frac{3}{2} I - \frac{1}{2} Z = \mathrm{diag}(1, 2), \qquad H_x = G \otimes X, \quad H_y = G \otimes Y, \quad H_q = X \otimes I. \tag{1}$$

The first factor is called the weight factor and the second a logical factor. We use the Hermitian Lie bracket

$$[\![A, B]\!] = \frac{[A, B]}{i}. \tag{2}$$

Theorem 2.1 (Exact weight-update closure)**.** *The real Hermitian Lie algebra generated by* (1) *has dimension ten and Pauli support*

$$\mathcal{B} = \{IX, ZX, IY, ZY, XI, IZ, ZZ, YX, YY, YZ\}. \tag{3}$$

It is centerless and perfect. For the anti-Hermitian representation $T = iH$, the space of antisymmetric forms satisfying

$$T^{\mathsf{T}}\mathcal{J} + \mathcal{J}T = 0 \tag{4}$$

is one-dimensional and contains the nonsingular witness $\mathcal{J} = iZ \otimes Y$, with $\det \mathcal{J} = 1$. The compact algebra is therefore

$$\mathfrak{g} \cong \mathfrak{usp}(4) \cong \mathfrak{so}(5). \tag{5}$$

Proof. Starting with H_x, H_y, H_q, repeatedly append every nonzero bracket that increases the exact flattened matrix rank. The process stops at the ten independent strings in (3). Exact structure constants give zero center and a ten-dimensional derived algebra; the Killing form in the computed real basis is negative definite. Solving the displayed linear equations for a general 4×4 antisymmetric matrix gives a one-dimensional null space and the stated nonsingular witness. Thus the defining module is symplectic, the algebra is compact and ten-dimensional, and the standard compact classification identifies it with $\mathfrak{usp}(4)$, equivalently $\mathfrak{so}(5)$. □

The classification is not inferred from dimension alone: the invariant form, compact Killing signature, center, and derived algebra are separately checked. Notice also what the theorem does not say. Finite-dimensional Hermitian evolution produces a compact unitary group; no noncompact Lorentzian real form follows by relabeling generators.

3 The weight as Cartan involution and trace selector

Introduce five Hermitian matrices

$$\Gamma_1 = Y \otimes I, \quad \Gamma_2 = X \otimes X, \quad \Gamma_3 = X \otimes Y, \quad \Gamma_4 = X \otimes Z, \quad \Gamma_5 = Z \otimes I \tag{6}$$

and bivectors

$$\Sigma_{AB} = \frac{[\Gamma_A, \Gamma_B]}{2i}, \qquad 1 \le A < B \le 5. \tag{7}$$

Proposition 3.1 (Clifford carrier)**.** *The matrices* (6) *obey*

$$\{\Gamma_A, \Gamma_B\} = 2\delta_{AB} I_4, \qquad \Gamma_1\Gamma_2\Gamma_3\Gamma_4\Gamma_5 = I_4. \tag{8}$$

The ten matrices Σ_{AB} have rank ten and span the algebra of Theorem 2.1.

Proof. Direct Pauli multiplication proves the anticommutators and ordered product. Flattening the ten bivectors gives matrix rank ten. Expressing them and (3) in the Pauli basis shows equality of the two spans. □

The traceless direction of the original weight is $S = \Gamma_5$. Define

$$\mathfrak{h} = \operatorname{span}\{\Sigma_{ab} : 1 \le a < b \le 4\}, \qquad \mathfrak{p} = \operatorname{span}\{\Sigma_{a5} : 1 \le a \le 4\}. \tag{9}$$

Theorem 3.2 (Intrinsic symmetric pair)**.** *The weight direction S has a six-dimensional centralizer in $\mathfrak{g}$ and grades it as*

$$\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{p}, \qquad [\mathfrak{h},\mathfrak{h}] \subset \mathfrak{h}, \quad [\mathfrak{h},\mathfrak{p}] \subset \mathfrak{p}, \quad [\mathfrak{p},\mathfrak{p}] \subset \mathfrak{h}. \tag{10}$$

Here $[S,\mathfrak{h}] = 0$ and $\{S,\mathfrak{p}\} = 0$. Hence the generating weight itself selects the compact symmetric pair $SO(5)/SO(4)$.

Proof. All assertions follow by exact multiplication of (6)–(7). The dimension of the centralizer is the nullity of the linear map $A \mapsto [A,S]$ restricted to the ten-dimensional span; it is six. □

The same S also selects an invariant tensor. Let $\epsilon_{1234} = 1$ and

$$\rho_\nu = \frac{1}{4}(I_4 + \nu S), \qquad -1 \le \nu \le 1. \tag{11}$$

Its eigenvalues are $(1 \pm \nu)/4$, each with multiplicity two.

Proposition 3.3 (Polarized trace identity)**.** *For $a,b,c,d \in \{1,2,3,4\}$,*

$$\mathrm{Tr}(S\Sigma_{ab}\Sigma_{cd}) = -4\epsilon_{abcd}, \tag{12}$$

$$\mathrm{Tr}(\rho_\nu\Sigma_{ab}\Sigma_{cd}) = \delta_{ac}\delta_{bd} - \delta_{ad}\delta_{bc} - \nu\epsilon_{abcd}. \tag{13}$$

The S-polarized mixed $\mathfrak{h}$–$\mathfrak{p}$ and $\mathfrak{p}$–$\mathfrak{p}$ traces vanish.

Proof. Equation (12) is the Clifford trace of four distinct coset directions with $S = \Gamma_5$ and the orientation fixed by the ordered product in Proposition 3.1. Expanding (11) gives (13); direct exact traces cover repeated indices and the vanishing sector statements. □

The epsilon contraction in (13) is the algebraic ingredient used in MacDowell–Mansouri-type broken curvature actions [6, 7]. Those actions are not derived here. The audited statement is narrower: one microscopic operator both generated the finite algebra and selected its symmetric pair and broken trace.

4 A two-cell bond and chiral Spin(10)

Consider two copies of $\mathcal{H}$. The uncoupled local algebra is generated by

$$L_{AB} = \Sigma_{AB} \otimes I_4, \qquad R_{AB} = I_4 \otimes \Sigma_{AB}. \tag{14}$$

Add the polarization bond $B = S \otimes S$. Its orbit under independent left and right local actions consists of the span

$$M_{A\dot{B}} = \Gamma_A \otimes \Gamma_{\dot{B}}, \qquad A,\dot{B} = 1,\dots,5. \tag{15}$$

Theorem 4.1 (Selective two-cell closure)**.** *The left, right, and cross subspaces have dimensions* 10*,* 10*, and* 25*, respectively, and satisfy*

$$[\mathfrak{g}_L,\mathfrak{g}_R] = 0, \qquad [\mathfrak{g}_L \oplus \mathfrak{g}_R,\mathfrak{m}] \subset \mathfrak{m}, \qquad [\mathfrak{m},\mathfrak{m}] \subset \mathfrak{g}_L \oplus \mathfrak{g}_R. \tag{16}$$

Both $S \otimes S$ and $\sum_A \Gamma_A \otimes \Gamma_A$ have a 25-dimensional local orbit. Either bond therefore generates a 45-dimensional compact algebra with decomposition

$$\mathfrak{spin}(10) = \big(\mathfrak{spin}(5),\mathbf{1}\big) \oplus \big(\mathbf{1},\mathfrak{spin}(5)\big) \oplus (\mathbf{5},\mathbf{5}). \tag{17}$$

Proof. Exact flattened ranks give $10 + 10 + 25 = 45$. Projecting every pairwise bracket onto the three subspaces gives zero residual. Successive adjoint action of the 20 local generators on either bond spans all 25 cross matrices. The bracket pattern, compactness, dimension, and standard symmetric decomposition identify the algebra as $\mathfrak{spin}(10)$; the decomposition itself is standard [8]. □

The paired Hilbert space has dimension 16. Its chirality can be read without assuming a branching table. Define

$$H_1 = \Sigma_{12} \otimes I, \quad H_2 = \Sigma_{34} \otimes I, \quad H_3 = I \otimes \Sigma_{12}, \quad H_4 = I \otimes \Sigma_{34}, \quad H_5 = S \otimes S. \tag{18}$$

Theorem 4.2 (Direct chirality certificate)**.** *The five H_k commute, obey $H_k^2 = I_{16}$, and satisfy $H_1H_2H_3H_4H_5 = I_{16}$. For $s \in \{\pm 1\}^5$, the joint projector*

$$P_s = \prod_{k=1}^{5} \frac{I_{16} + s_k H_k}{2} \tag{19}$$

has rank one if $\prod_k s_k = +1$ and rank zero if $\prod_k s_k = -1$. The commutant of the local product algebra is scalar. Thus the paired space is one irreducible chiral 16 of Spin(10)*.*

Proof. The commuting and product identities follow exactly from Proposition 3.1. Expanding all 32 projectors gives the stated ranks. Solving the commutant equations for a general 16×16 matrix gives dimension one, proving irreducibility. □

This closure is selective. A bivector dot product, a rotation-only dot product, or a coset-only dot product generates the full $\mathfrak{su}(16)$ under independent local actions, whereas a single vector bridge and the two bonds in Theorem 4.1 generate the 45-dimensional algebra. Selection of the bond is therefore substantive; it is not a generic property of two four-level systems.

5 A positive-weight family and an essential control restriction

The rational weight above is not an isolated point. The following statement identifies which part of the construction survives a change of eigenvalues.

Theorem 5.1 (Positive-weight family)**.** *Let $G = aI + bZ$, where $a > |b| > 0$, and allow the three controls $G \otimes X$, $G \otimes Y$ and $X \otimes I$ independently. Their real Hermitian Lie algebra is the same ten-dimensional span as in Eq.* (3)*.*

Proof. With $Q = X \otimes I$, direct multiplication gives

$$[\![Q, [\![Q, G \otimes X]\!]]\!] = -4bZ \otimes X, \qquad [\![Q, [\![Q, G \otimes Y]\!]]\!] = -4bZ \otimes Y. \tag{20}$$

Since $a, b \neq 0$, linear combinations recover IX, IY, ZX, ZY. Their brackets, together with XI, generate IZ, ZZ, YX, YY, YZ. All three original controls belong to the ten-string span, which is closed under the bracket, so there are neither fewer nor more generators. □

Positivity does not force this closure for an arbitrary update that fails to commute with G. For example, keeping $G = \mathrm{diag}(1, 2)$ and replacing XI by $(X + Z) \otimes I$ produces dimension 15, hence all of $\mathfrak{su}(4)$. The accompanying exact matrix algorithm starts from those three controls and repeatedly adds a bracket with an original control only when its flattened rank increases. It terminates with the full fifteen-dimensional traceless Hermitian space. The fixed-point span is invariant under the original controls and therefore, by Jacobi, under their generated algebra; thus termination is an exact finite certificate. This counterexample limits the mechanism to the stated transverse-update family. The direction S is encoded in the control coefficients, but it is not itself one of the ten Lie generators. The state parameter ν in Eq. (11) remains additional information; a direction selected by a weight does not fix a polarization magnitude or an action coefficient.

6 Exact multicell proliferation

The two-cell polarization bond is unusually small: independent local $\mathfrak{spin}(5)$ actions plus S_1S_2 close on $\mathfrak{spin}(10)$. We now test whether repeating the bond produces a spatial chain. In the Pauli frame used above, one cell has the ten-string basis

$$\{IX, ZX, IY, ZY, XI, IZ, ZZ, YX, YY, YZ\}, \tag{21}$$

with polarization $S = ZI$ and clock bivector $C = YZ$.

Every n-qubit Pauli string is represented by a binary vector $(x|z) \in \mathbb{F}_2^{2n}$. Two strings anticommute precisely when their symplectic product is one; their nonzero commutator is proportional to the string encoded by their binary sum. Closing a set under this rule is therefore exact and avoids numerical matrix ranks. This binary representation is standard in Pauli-algebra analysis and quantum control [9].

Theorem 6.1 (Three-cell proliferation obstruction)**.** *Allow the ten local generators* (21) *independently on each cell of an open chain. Add nearest-neighbor bonds of either type S_rS_{r+1} or C_rC_{r+1}. The exact Lie-closure dimensions are*

cells	S_rS_{r+1}	C_rC_{r+1}	$\dim \mathfrak{su}(4^n)$
1	*10*	*10*	*15*
2	*45*	*255*	*255*
3	*4095*	*4095*	*4095*

Thus three independently controlled cells with either raw bond generate all of $\mathfrak{su}(64)$. In particular, the two-cell $\mathfrak{spin}(10)$ closure does not iterate as a chain law.

Proof. Encode all local strings and bonds as binary symplectic vectors. Repeatedly add the binary sum of every anticommuting pair until no new vector appears. The exact cardinalities are those in the table. For three cells the cardinality $4095 = 4^6 - 1$ is the number of all nonidentity Pauli strings on six qubits, which span $\mathfrak{su}(64)$ as Hermitian generators. □

The theorem rules out a tempting interpretation: the paired Spin(10) carrier cannot simply be placed on every spatial link while retaining independent cell controls. It may still serve as an on-site representation, but a scalable low-energy sector would need a symmetry, a gap, or a projection that changes the effective algebra.

7 Classification of the specified connected networks

The three-cell computation has an exact extension to every larger connected graph. Network-controllability arguments and algebraic propagation are established tools [10]; the point here is to close the extension for the two specified bonds and the restricted local algebra.

Lemma 7.1 (Extension of a fully controlled block)**.** *Suppose a block of $m \geq 2$ cells generates $\mathfrak{su}(d)$, where $d = 4^m$. Append a four-level cell with its independent local $\mathfrak{spin}(5)$ controls. If a connecting control is $A_0 \otimes \Gamma_5$ or $A_0 \otimes \Sigma_{45}$ with A_0 a nonzero traceless Hermitian block operator, the enlarged control algebra is $\mathfrak{su}(4d)$.*

Proof. Write $V = \mathrm{span}_{\mathbb{R}}\{\Gamma_i\}$ and $B = \mathrm{span}_{\mathbb{R}}\{\Sigma_{ij}\}$. The real adjoint module of the compact simple algebra $\mathfrak{su}(d)$ is irreducible. The local V and B modules are respectively the irreducible vector and adjoint modules of $\mathfrak{spin}(5)$. Independent repeated commutation with block and local controls therefore generates all of $\mathfrak{su}(d) \otimes V$ from a vector bond, or all of $\mathfrak{su}(d) \otimes B$ from a bivector bond.

In the vector case choose two commuting block Pauli involutions A, B_0 whose product is nonzero and traceless, such as two independent Z strings. Then

$$[\![A \otimes \Gamma_1, B_0 \otimes \Gamma_2]\!] = 2AB_0 \otimes \Sigma_{12}. \tag{22}$$

Taking the block adjoint span and the local bivector orbit gives $\mathfrak{su}(d) \otimes B$ too. Conversely, in the bivector case choose anticommuting block Pauli involutions A, B_0 and the commuting local bivectors Σ_{12}, Σ_{34}. The Clifford volume convention gives $\Sigma_{12}\Sigma_{34} = -\Gamma_5$, hence

$$[\![A \otimes \Sigma_{12}, B_0 \otimes \Sigma_{34}]\!] = -[\![A, B_0]\!] \otimes \Gamma_5. \tag{23}$$

This generates the vector module as well.

Choose a traceless block involution Q with $Q^2 = I$. Since both tensor modules are available,

$$[\![Q \otimes \Sigma_{12}, Q \otimes \Gamma_1]\!] = 2I \otimes \Gamma_2. \tag{24}$$

Local rotations supply every $I \otimes \Gamma_i$. Together with the original $I \otimes B$, these form $I \otimes \mathfrak{su}(4)$, because the five vectors and ten bivectors span all traceless Hermitian 4×4 matrices. Independent block and full local adjoint actions on any nonzero tensor then supply $\mathfrak{su}(d) \otimes \mathfrak{su}(4)$. The three resulting subspaces have total dimension

$$(d^2 - 1) + 15 + 15(d^2 - 1) = 16d^2 - 1, \tag{25}$$

which exhausts $\mathfrak{su}(4d)$. □

Theorem 7.2 (Connected-network control algebra)**.** *On a finite connected graph, allow independent copies of the ten local generators and independently available nonzero edge controls of type S_rS_s or C_rC_s. One cell generates $\mathfrak{spin}(5)$. Two cells with only an S bond generate $\mathfrak{spin}(10)$; with a C bond present they generate $\mathfrak{su}(16)$. Every connected graph of $n \geq 3$ cells with these edge types generates $\mathfrak{su}(4^n)$.*

Proof. The one-, two- and three-cell bases are the exact Pauli closures in Theorem 6.1. If a C edge is present, its two endpoints already give a fully controlled $\mathfrak{su}(16)$ block. If every edge is of type S, a connected graph with at least three vertices contains a three-vertex path, whose controls generate $\mathfrak{su}(64)$. In either case grow the controlled block along a spanning tree. At each step the block part of the next edge is a nonzero traceless operator, so Lemma 7.1 applies. Extra available edges cannot reduce a generated algebra. □

The assumption of independently available controls matters. A single fixed homogeneous Hamiltonian, a diagonal collective control restriction or a projected low-energy model is a different problem. The theorem rules out the naive repetition of this pair as a restricted spatial control algebra; it does not rule out every constrained or collective model. As a portability check, the integer algorithm also gives $65535 = \dim \mathfrak{su}(256)$ for four-cell chains with either uniform bond. This check supports the implementation; the analytic induction, rather than an extrapolation of dimensions, proves the all-graph result.

8 The nested $SO(3)$ decomposition

Using the Clifford matrices and bivectors already fixed above, set

$$J_i = \frac{1}{2}\epsilon_i{}^{jk}\Sigma_{jk}, \qquad K_i = \Sigma_{i4}, \qquad E_i = \Sigma_{i5}, \qquad C = \Sigma_{45}, \tag{26}$$

where $i, j, k \in \{1, 2, 3\}$. We again use $[\![A, B]\!] = [A, B]/i$.

Proposition 8.1 (Exact finite phase-space algebra)**.** *The ten operators in* (26) *are independent and decompose as*

$$\mathbf{10} = \mathbf{3}_J \oplus \mathbf{3}_K \oplus \mathbf{3}_E \oplus \mathbf{1}_C \tag{27}$$

under the rotations generated by J_i. Their nonzero brackets are

$$\begin{aligned}
[\![J_i, J_j]\!] &= 2\epsilon_{ij}{}^k J_k, & [\![J_i, K_j]\!] &= 2\epsilon_{ij}{}^k K_k, & [\![J_i, E_j]\!] &= 2\epsilon_{ij}{}^k E_k, \\
[\![K_i, K_j]\!] &= 2\epsilon_{ij}{}^k J_k, & [\![E_i, E_j]\!] &= 2\epsilon_{ij}{}^k J_k, & [\![K_i, E_j]\!] &= 2\delta_{ij} C, \\
[\![C, K_i]\!] &= 2E_i, & [\![C, E_i]\!] &= -2K_i, & [\![C, J_i]\!] &= 0.
\end{aligned} \tag{28}$$

Proof. Exact flattened rank is ten. Substitution of the Clifford matrices and Pauli multiplication yields every bracket in (28). The J brackets then display two vectors and one scalar under the common $SO(3)$. □

This is a compact Yang/Snyder-type phase-space algebra, not a new abstract algebra. Our use of it is tied to the specific cell: E_i are simultaneously part of its weight-selected Clifford coset, and C anticommutes with the weight polarization introduced next.

9 Double polarization and the quantum central limit

Let

$$S = \Gamma_5, \qquad \rho_{\nu,\mu} = \frac{1}{4}(I + \nu S + \mu C). \tag{29}$$

Since $S^2 = C^2 = I$ and $\{S, C\} = 0$, this two-parameter state can be analyzed exactly.

Theorem 9.1 (Positive double-polarized cell)**.** *The spectrum of* (29) *is*

$$\frac{1 \pm \sqrt{\nu^2 + \mu^2}}{4}, \tag{30}$$

each eigenvalue occurring twice. Hence $\rho_{\nu,\mu} \geq 0$ exactly when $\nu^2 + \mu^2 \leq 1$. For generic nonzero ν, μ, the common centralizer of S and C inside $\mathfrak{spin}(5)$ is the three-dimensional algebra generated by J_i. The one-cell moments are

$$\begin{aligned}
\langle C \rangle &= \mu, & \langle J_i \rangle = \langle E_i \rangle = \langle K_i \rangle &= 0, \\
\frac{1}{2}\langle \{E_i, E_j\} \rangle &= \delta_{ij}, & \frac{1}{2}\langle \{K_i, K_j\} \rangle &= \delta_{ij}, \qquad (31) \\
\langle [\![K_i, E_j]\!] \rangle &= 2\mu\delta_{ij}. & & \qquad (32)
\end{aligned}$$

Proof. The anticommuting involutions give $(\nu S + \mu C)^2 = (\nu^2 + \mu^2)I$, proving the spectrum and positivity disk. The centralizer is the null space of the two simultaneous adjoint maps restricted to the ten-dimensional algebra and has dimension three. Direct traces of (29) with the matrices in (26) give (31). □

For N independent cells in the product state $\rho_{\nu,\mu}^{\otimes N}$, define the fluctuation operator

$$F_N(A) = \frac{1}{\sqrt{N}} \sum_{r=1}^{N} \big(A_r - \langle A \rangle I\big). \tag{33}$$

Assume $\mu > 0$ and set

$$e_i^{(N)} = \frac{F_N(E_i)}{\sqrt{2\mu}}, \qquad k_i^{(N)} = \frac{F_N(K_i)}{\sqrt{2\mu}}. \tag{34}$$

Theorem 9.2 (Three-pair CCR contraction)**.** *In the product-state quantum central limit, the six variables in* (34) *converge in moments to a regular CCR algebra with*

$$[\![k_i, e_j]\!] = \delta_{ij}, \qquad [\![e_i, e_j]\!] = [\![k_i, k_j]\!] = 0. \tag{35}$$

Their symmetric covariance is diagonal and

$$\mathrm{Var}(e_i) = \mathrm{Var}(k_i) = \frac{1}{2\mu}. \tag{36}$$

Consequently positivity imposes the kinematic bound

$$\mathrm{Var}(e_i) = \mathrm{Var}(k_i) \geq \frac{1}{2\sqrt{1-\nu^2}} \tag{37}$$

at fixed ν and $\mu > 0$.

Proof. The product-state noncommutative central-limit theorem maps centered one-site observables to a CCR fluctuation algebra whose symplectic form is the expectation of their commutator [11]. Equation (31), together with the normalization in (34), gives (35) and (36). Maximizing μ on the positivity disk at fixed ν gives the last inequality. □

At finite N the same scaling gives

$$\left[\!\!\left[k_i^{(N)}, e_j^{(N)}\right]\!\!\right] = \delta_{ij}\frac{C^{(N)}}{N\mu}, \qquad \left[\!\!\left[e_i^{(N)}, e_j^{(N)}\right]\!\!\right], \ \left[\!\!\left[k_i^{(N)}, k_j^{(N)}\right]\!\!\right] = O\left(\frac{J^{(N)}}{N}\right). \tag{38}$$

The contraction is therefore visible directly in the finite algebra when $C^{(N)}/N \to \mu$ and $J_i^{(N)}/N \to 0$ in the chosen state. Calling (e, k) "triad" and "extrinsic curvature" would be only a proposed interpretation: no spatial index, derivative, or Hamiltonian constraint has yet been constructed.

10 A dynamical origin for the polarized state

The state (29) need not be inserted by hand. Define collective sums $S_N = \sum_r S_r$ and $C_N = \sum_r C_r$ and the permutation-invariant Hamiltonian

$$H_N^{\mathrm{pol}} = -\frac{J}{2N}S_N^2 - hC_N, \qquad J, h > 0. \tag{39}$$

The thermodynamic-limit mean-field functional for the order parameter $\nu = \langle S_N\rangle/N$ is

$$f(\nu) = \frac{J}{2}\nu^2 - \frac{1}{\beta}\log\Big[4\cosh\Big(\beta\sqrt{J^2\nu^2 + h^2}\Big)\Big]. \tag{40}$$

Such thermodynamic limits and variational descriptions are standard for mean-field spin and quantum Curie–Weiss models [12, 13].

Theorem 10.1 (Mean-field double-polarization phase)**.** *Let $r = \sqrt{J^2\nu^2 + h^2}$. The derivatives and self-consistency equations are*

$$f'(\nu) = J\nu\left[1 - \frac{J}{r}\tanh(\beta r)\right], \tag{41}$$

$$f''(0) = \frac{J}{h}\left[h - J\tanh(\beta h)\right], \tag{42}$$

$$\nu = \frac{J\nu}{r}\tanh(\beta r), \qquad \mu = \frac{h}{r}\tanh(\beta r). \tag{43}$$

The continuous mean-field boundary is

$$J\tanh(\beta h) = h, \tag{44}$$

with an ordered solution when the left side exceeds h. At zero temperature and $J > h$, the two symmetry-related ordered states have

$$\nu = \pm\sqrt{1-(h/J)^2}, \qquad \mu = h/J. \tag{45}$$

At finite temperature, $\sqrt{\nu^2+\mu^2} = \tanh(\beta r) < 1$.

Proof. Differentiate (40) exactly to obtain (41)–(42). The one-site effective Hamiltonian is $-J\nu S - hC$; because S and C anticommute, its Gibbs state is of the form (29) and yields (43). In the zero-temperature ordered phase, nonzero ν implies $r = J$, giving (45). □

Equation (39) is spatially ultralocal: it explains the internal block phase, not curvature or transport. The mean-field result also does not prove stability under finite-range interactions in three dimensions.

11 The topology of the collective limit

The state-dependent contraction has a useful exact error estimate. It also has a sharp obstruction to an interpretation as microscopic operator convergence.

Proposition 11.1 (State error and operator-norm obstruction)**.** *For fixed $\mu > 0$ in the positivity disk and the iid product state, let*

$$R_N = \left[\!\!\left[k_i^{(N)}, e_i^{(N)}\right]\!\!\right] - I = \frac{C_N}{N\mu} - I. \tag{46}$$

Then

$$\langle R_N^2\rangle = \frac{1-\mu^2}{N\mu^2}, \qquad \|R_N\| = 1 + \frac{1}{\mu}. \tag{47}$$

Proof. The one-cell identities $C^2 = I$ and $\langle C\rangle = \mu$, together with independence, give $\mathrm{Var}(C_N) = N(1-\mu^2)$ and $\langle C_N\rangle = N\mu$. This proves the first identity. The spectrum of C_N/N includes both -1 and $+1$ for every N. Since $0 < \mu \le 1$, the largest absolute eigenvalue of R_N is $1 + 1/\mu$. □

The product-state root-mean-square error is therefore $\sqrt{1-\mu^2}/(\mu\sqrt{N})$, while the full operator norm never tends to zero. At $\mu = 1$ the state error is zero but the full norm is still two. At $\mu = 0$ the stated normalization is undefined and the unnormalized E, K fluctuation symplectic form is degenerate. For $\mu < 0$ one may normalize with $|\mu|$ and reverse one canonical orientation, but a negative variance $1/(2\mu)$ is not admissible. If μ varies with N and tends to zero, even commutator convergence in state root mean square requires $N\mu_N^2 \to \infty$; the canonical variances still diverge, so this condition alone does not give the fixed regular Gaussian limit used here.

The product-state hypothesis can be seen directly in the moment proof. On expanding a centered product of fluctuation sums, a site appearing only once contributes zero. After the $N^{-k/2}$ normalization, only partitions into pairs survive for even k; odd moments vanish. The ordered two-point functions determine a Gaussian noncommutative state, whose symmetric part is the covariance and whose antisymmetric part is the expected commutator. The bounded finite-dimensional one-site observables make all these moments finite. The limiting covariance satisfies the CCR positivity condition since $0 < \mu \le 1$, and its Gaussian characteristic functions are continuous, giving a regular CCR state. This argument also permits nonfaithful one-site states on the positivity-disk boundary when $\mu > 0$.

12 What the phase Hamiltonian does establish

The phase model above has an exact reduction to established mean-field physics. The two anticommuting involutions S, C generate an associative algebra unitarily equivalent to $M_2(\mathbb{C}) \otimes I_2$. To see this, diagonalize S into two eigenspaces of dimension two; C interchanges them unitarily, and a choice of basis in one eigenspace makes its off-diagonal block the identity. A further two-dimensional rotation maps the pair to $Z \otimes I_2, X \otimes I_2$. Thus the block Hamiltonian is precisely a transverse-field spin-half Curie–Weiss model with a free two-dimensional spectator at each site. The spectator multiplies the partition function by 2^N and shifts the free-energy density by $-\log(2)/\beta$. This explains the factor $4\cosh$ rather than $2\cosh$ and identifies the critical line with the established $p = 2$ result [13]. No new mean-field phase boundary is claimed.

The mean-field variational problem supplies an extremal one-site equilibrium with the required polarizations. It does not prove that the actual interacting finite-N Gibbs states have the same fluctuation covariance as the iid state. Finite Gibbs states preserve the $S \mapsto -S$ symmetry, so selecting one ordered sign requires a symmetry-breaking limit. Moreover, correlations invisible in a fixed number of one-site marginals can survive $N^{-1/2}$ fluctuation scaling. The product-state CCR theorem and the phase theorem therefore establish compatible one-site data, not a theorem identifying their full collective fluctuation states. This distinction is needed before claiming that the Hamiltonian dynamically realizes the exact noise formula.

13 What collective symmetry protects—and what it does not

Every polynomial in the collective sums of a block commutes with the action of all within-block permutations. Hence each permutation sector is exactly invariant under (39) and under any collective-polynomial extension.

Proposition 13.1 (Exact permutation-sector protection)**.** *Let P_λ project onto any isotypic sector of the permutation action on $(\mathbb{C}^4)^{\otimes N}$. If H is a polynomial in collective sums $\sum_r A_r$, then*

$$[H, P_\lambda] = 0, \qquad (I - P_\lambda) H P_\lambda = 0. \tag{48}$$

Proof. Each collective sum is invariant under permutation conjugation. So is every polynomial in such sums. The invariant operator is block diagonal in the permutation-isotypic decomposition, proving both identities. □

This exact statement is weaker than gravitational protection. Suppose P_N denotes a proposed low-energy/constraint projector and $\widehat{C}_I^{(N,a)}$ microscopic constraints at cell number N and spatial resolution a. A viable continuation must establish both

$$\left\| P_N \left(\left[\!\left[\widehat{C}_I, \widehat{C}_J \right]\!\right] - \widehat{f}_{IJ}{}^K \widehat{C}_K \right) P_N \right\| \le \epsilon_{N,a}, \qquad \epsilon_{N,a} \longrightarrow 0, \tag{49}$$

$$\left\| (I - P_N) \widehat{C}_I P_N \right\| \longrightarrow 0. \tag{50}$$

Product-state fluctuation estimates suggest $O(N^{-1/2})$ kinematic corrections, while a smooth centered spatial difference is often $O(a^2)$. Balancing them would suggest $N^{-1/2} = o(a^2)$, for example $N \gg a^{-4}$. This scaling is a heuristic target, not a theorem: inverse operators, correlations, and nonsmooth states may invalidate it.

14 Interpretation and reproducibility

The positive-weight family gives an economical finite control realization of familiar compact structures. Its symmetric pair and epsilon trace are the algebraic ingredients used in known

broken-curvature constructions [6, 7]; neither an action nor a Lorentzian real form follows from the control algebra. The paired chiral representation is likewise a representation result, not a derivation of fermion statistics or a field theory. The all-network theorem makes the principal obstruction exact: unrestricted local controls and these raw bonds cannot define a spatial network whose algebra remains the exceptional paired algebra. A constrained model, restricted control set, symmetry sector or low-energy projection would require additional hypotheses and a separate analysis.

The iid collective theorem is a valid alternative kinematic construction. It has three nondegenerate CCR pairs for fixed positive clock polarization, but it supplies no spatial derivative, Hamiltonian constraint, gravitational projector or continuum propagation. Exact permutation-sector protection remains weaker than protection of an intended physical constraint sector. In particular, Proposition 11.1 prevents an $N^{-1/2}$ state-fluctuation estimate from being presented as the operator-norm closure bound written above. The heuristic spatial double scaling is not a consequence of the results proved here.

The portable script `code/verify.py` reconstructs the one-cell closure with exact matrices, checks the invariant symplectic form, Clifford and polarized trace identities, all thirty-two paired Cartan projectors, the six-variable moments, the displayed brackets, the mean-field derivatives and the update counterexample. It also performs integer symplectic closure for one through four cells with both bonds. All output is written under `outputs/`; the rational Pauli definitions and algorithm are included in the script and `inputs/model.json`. Closing a Pauli set under repeated adjoint actions of the original generators suffices by Jacobi; every nonzero bracket is the binary XOR of two anticommuting strings. The independently reproduced dimensions are $(10, 45, 4095, 65535)$ for vector bonds and $(10, 255, 4095, 65535)$ for clock bonds. The analytic induction establishes the arbitrary-network theorem; no finite table is extrapolated to prove it.

The supplementary verification record separates new exact checks, reused historical results and unproved physical extensions. The finite Lie mechanisms and central-limit tools have substantial prior literature [9, 10, 11]; the particular construction and its control boundary are the focus, and no exhaustive publication-priority claim is made. The current exact checks do not replay every non-Pauli bond scan or establish the fluctuation law of the actual interacting Gibbs state.

Contribution history and authorship status

Human author names and affiliations are intentionally blank, and human expert review has not occurred. Earlier work was Fable 5-led; results from July 2026 onward were predominantly produced by GPT models, as reported by the operator. DeepSeek and multiple GPT models also participated in review and preparation. The present preparation combines the finite-cell and collective-network results and adds the positive-weight family, explicit update counterexample, network induction and exact convergence-topology boundary. Standard Clifford theory, control-algebra principles, quantum central limits and Curie–Weiss thermodynamics are credited rather than attributed as new results. The stated chronology does not resolve authorship or intellectual ownership:

> I genuinely do not know how authorship, intellectual credit, or ownership should be attributed for these results.

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